

DARPA RSGS Space Mech

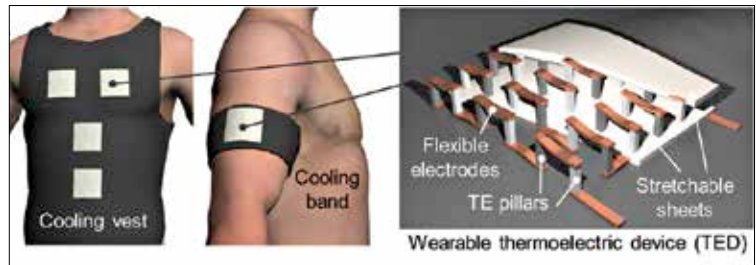
DARPA, the Defense Advanced Research Projects Agency is working on a new high-tech spacecraft. Which is armed. Not the kind of armed you're thinking about. It's literally armed, as in, it has arms. You know, the kind we use to grab things.

Mechanical arms in themselves aren't exactly a new concept, they have been around for a while now, and we've used them just about everywhere – factories, hospitals to perform surgeries, undersea exploration vehicles and more – space however, is definitely a whole new frontier.



Now the reason DARPA is making a space mech is for satellite maintenance. It makes sense, satellites are expensive, they can cost billions of dollars to get into orbit, but once they're there, we just cross our fingers and hope everything will stay okay and in place. Now hopefully, if something goes wrong, DARPA's space mech can do something about it. The RSGS (Robotic Servicing of Geosynchronous Satellites) is like a commercial satellite that has two arms on it. These arms come equipped with interchangeable tools, allowing for a variety of different operations. These include being able to latch onto the spacecraft and commence repairs.

If things go well, with the RSGS, it would be the first time in human history that it would be possible to carry out tasks to help fix satellites in orbit. Until it's needed however, the RSGS will just chill in orbit.



Adaptive thermo-sensor camouflage

A team of scientists at the University of California, San Diego have created a wearable thermo-camo which can fool heat sensors. Heat-detecting sensors, such as the kind in night vision goggles for example, will not be able to detect this wearable.

This is the first wearable of its kind, which allows users to adjust their own temperature. Of course, it's only the wearer's body heat that's camouflaged, the wearer themselves are unaffected on the inside and remain at a comfy temperature.

The wearable can go between 10 to 38 degrees celsius in under a minute. The device is also wireless and can be embedded into fabric, within an armband for instance. They're working on a more advanced version of the wearable which can fit inside jackets and more. It already sounds like a must have device for spies and secret agents.

The team's current challenge is scaling up the tech, while right now it can only work at the armband scale, they hope to eventually make an entire garment embedded with the technology. As it stands right now though, the garment would end up being too heavy and would only work for a total of one hour, so work is still in progress.

Everest customisable keyboard

There's no dearth of gaming keyboards out there. Mountain.gg have come up with what they think should be your one-stop-shop for everything gaming keyboard with the Everest keyboard. This is a fully modular mechanical keyboard. It comes with a wide range of customisable features as well. The modular options? Just about everything apparently. You can remove the numpad and the media dock, switch them around, or even move them to the left side. The keycaps are all changeable, and so are the mechanical switches. The detachable numpad also includes four fully customisable display keys. You can programme your favourite short-



cuts or icons to these keys, be it your favourite game or software. The detachable media dock also comes with a display dial, which is ALSO modular and can be moved to either corner of the media dock. You can have your pick of Cherry MX switches, and you can even hot-swap the switches as long as they're compatible with plate mounted switches. No need to worry about soldering.